

Linear system: $L[au + bv] = aL[u] + bL[v]$, $a, b \in \mathbb{R}$

LTI: $y(t) = L[x(t)] \implies y(t - t_0) = Lx(t - t_0)$

Delta function:

1. $\int \delta(t) dt = 1$
2. $\delta(t) = 0$ for $t \neq 0$
3. $\int f(t)\delta(t - t_0) dt = f(t_0)$ (sifting)

Impulse response: the output of an LTI system when the input is $\delta(t)$

Transfer function: $H(f) = \mathcal{F}(h(t))$ where $h(t)$ is the impulse response of the system.

$$y(t) = (x * h)(t) = \int_{-\infty}^{\infty} x(\tau)h(t - \tau) d\tau$$

(in fact, in the frequency domain, this operation is defined by $Y(f) = X(f)H(f)$)

Fourier Transform:

$$\mathcal{F}[x(t)] = \int x(t)e^{-j2\pi ft} dt = X(f)$$

Inverse Fourier:

$$\mathcal{F}^{-1}[X(f)] = \int X(f)e^{j2\pi ft} df = x(t)$$

Standard Fourier pairs:

1. $\mathcal{F}[\delta(t)] = 1$
2. $\mathcal{F}[e^{-j2\pi f_0 t}] = \delta(f + f_0)$
3. $\mathcal{F}[\delta(t - t_0)] = e^{-j2\pi f t_0}$
4. $X(f - f_0) = x(t)e^{j2\pi f_0 t}$
5. $\mathcal{F}[\cos 2\pi f_0 t] = \frac{1}{2}[\delta(f + f_0) + \delta(f - f_0)]$
6. $\mathcal{F}[\sin 2\pi f_0 t] = \frac{1}{2j}[\delta(f + f_0) - \delta(f - f_0)]$
7. $\mathcal{F}[\mathbb{1}_{x \in [-T/2, T/2]}] = \text{sinc}(2\pi T f)$

Fourier Properties:

1. $\mathcal{F}[x(t - t_0)] = e^{-j2\pi f t_0} X(f)$
2. $\mathcal{F}[x(at)] = \frac{1}{|a|} X(f/a)$
3. $\mathcal{F}[x'(t)] = X(f) \cdot j2\pi f$
4. $\mathcal{F}[\int_{-\infty}^t x(u) du] = \frac{X(f)}{j2\pi f} + \frac{X(0)}{2} \delta(f)$
5. $\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$ (Parseval's)
6. $\mathcal{F}[x(-t)] = X(-f)$
7. $\mathcal{F}[X(t)] = x(-f)$

Useful tricks:

1. $|X(f)|^2 = X(f)X^*(f) = X(f)X(-f)$
2. $\cos(2\pi ft) = (e^{j2\pi ft} + e^{-j2\pi ft})/2$

$$3. \sin(2\pi ft) = (e^{j2\pi ft} - e^{-j2\pi ft})/2j$$

4. Fourier of odd function is purely imaginary, Fourier of even function is purely real.

5. δ is an even function: $\delta(-t) = \delta(t)$

$$6. \sin(x + y) = \sin x \cos y + \cos x \sin y$$

$$7. \cos(x + y) = \cos x \cos y - \sin x \sin y$$

$$8. \sin^2 x = (1 - \cos 2x)/2$$

$$9. \cos^2 x = (1 + \cos 2x)/2$$

$$10. \cos \theta(t) = \sum_{m=0}^{\infty} \frac{(\theta(t))^{2m}}{(2m)!}$$

$$11. \sin \theta(t) = \sum_{m=0}^{\infty} (-1)^m \frac{(\theta(t))^{2m+1}}{(2m+1)!}$$

$$12. \mathcal{F}[x(t)y(t)] = X(f) * Y(f)$$

Baseband: unmodulated signal (i.e. the message signal).

Passband: modulated signal (i.e. the transmitted signal).

AM Coherent modulation: $r(t) = m(t) \cos 2\pi f_c t$, demodulate with $\cos 2\pi f_c t$ and LPF to recover $m(t)/2$.

AM Large carrier: $r(t) = (m(t) + A) \cos 2\pi f_c t$ ($A \gg m(t)$), demodulate with an envelope detector, LPF, and HPF to recover $m(t)$.

AM Single sideband: $r(t) = m(t) \cos 2\pi f_c t$ with an LPF applied to $m(t)$ before modulation (takes advantage of symmetry from $M(-f) = M^*(f)$ to use half the bandwidth), demodulate with $\cos 2\pi f_c t$ and LPF to recover $m(t)/2$.

Frequency modulation: $r(t) = \sin(2\pi f_c t + \beta \int_{-\infty}^t m(\tau) d\tau)$, demodulate by differentiating, envelope detecting, and HPF to recover $\beta m(t)$.

Phase modulation: $r(t) = \sin(2\pi f_c t + \beta m(t))$ (filter and integrate)

Carson's Rule: $\Delta = (\kappa + 1)B$ where Δ is the single-sided FM bandwidth, κ is the modulation index, and B is the bandwidth of the message signal.

Sampling:

$$\begin{aligned} \hat{m}(t) &= \sum_k m(Tk) \delta(t - Tk) * \frac{\sin(\pi t/T)}{\pi t} \\ &= \sum_k m(Tk) \frac{\sin(\pi(t - Tk)/T)}{\pi(t - Tk)/T} \end{aligned}$$

corresponding to adding copies of the spectrum at $[k/T - B, k/T + B]$ for $k \in \mathbb{Z}$ via delta functions and then low pass filtering.

Low pass filter:

$$\begin{aligned} H(f) &= u(f + B) - u(f - B) \\ h(t) &= \mathcal{F}^{-1}[H(f)] = \int_{-B}^B e^{-j2\pi ft} df \\ &= \frac{e^{-j2\pi Bt} - e^{j2\pi Bt}}{-j2\pi t} \\ &= \frac{\sin(2\pi Bt)}{\pi t} = \frac{\sin(\pi t/T)}{\pi t} \end{aligned}$$

Nyquist Criterion: $T < 1/2B$ (i.e. sample at rate T to avoid the sampled spectra from overlapping)

- $m(t) \cos 2\pi t$ has bandwidth $B = 1$
- $m(t) \cos 2\pi f_c t$ has frequency in $[f_c - B, f_c + B]$ so we need $T < 1/2(f_c + B)$

Expected quantization error:

$$\int f_M(m)(m - Q(m))^2 dm$$

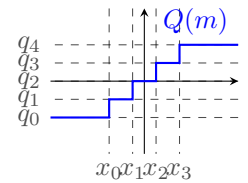
Leibniz's Rule:

$$\begin{aligned} \frac{d}{dx} \int_{a(x)}^{b(x)} f(x, y) dy &= \\ f(x, b(x)) \cdot \frac{db}{dx} - f(x, a(x)) \cdot \frac{da}{dx} \\ + \int_{a(x)}^{b(x)} \frac{\partial f}{\partial x} dy \end{aligned}$$

Lloyd max optimal quantizer:

1. $x_i = \frac{1}{2}(q_{i+1} + q_i)$
2. $q_i = \mathbb{E}[M \mid m \in (x_{i-1}, x_i)]$

(should be symmetric and have $N = 2^b$ quantization levels for b bits)



Convexity: $f(x_1, \dots, x_n)$ convex iff $\forall x_1, x_2 \in \mathbb{R}^n, \lambda \in [0, 1]$,

$$f(\lambda x_1 + (1 - \lambda)x_2) \leq \lambda f(x_1) + (1 - \lambda)f(x_2)$$

Hessian:

$$H_f(x) = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

f is convex iff $H_f(x)$ is positive semidefinite for all x : $x^T H_f(x) x \geq 0$ (equivalently all eigenvalues of $H_f(x)$ are nonnegative for all x)

PMF: $f_X(x) = \mathbb{P}(X = x)$ with $\int f_X(x) dx = 1$ and $f_X(x) \geq 0$ for all x .

CDF: $F_X(x) = \mathbb{P}(X \leq x) = \int_{-\infty}^x f_X(t) dt$

Expectation: $\mathbb{E}[X] = \int x f_X(x) dx$

Variance: $\text{Var}[X] = \mathbb{E}[(X - \mathbb{E}[X])^2] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2 = \sigma_X^2$

Joint random variables:

1. $\int \int f_{X,Y}(x, y) dx dy = 1$
2. $f_{X,Y}(x, y) \geq 0$

Marginals:

$$f_X(x) = \int f_{X,Y}(x, y) dy$$

$$f_Y(y) = \int f_{X,Y}(x, y) dx$$

Marginal Expectation:

$$\mathbb{E}[X] = \int \int x f_{X,Y}(x, y) dx dy$$

Conditional probability:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x, y)}{f_Y(y)}$$

Bayes: $f_{X,Y}(x, y) = f_{X|Y}(x|y)f_Y(y) = f_{Y|X}(y|x)f_X(x)$

Independence: $f_{X|Y}(x|y) = f_X(x)$

Correlation:

$$\mathbb{E}[XY] = \int \int xy f_{X,Y}(x, y) dx dy$$

Covariance: $\text{Cov}[X, Y] = \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \int \int (X - \bar{X})(Y - \bar{Y}) f_{X,Y}(x, y) dx dy$

Trick: If $\mathbb{E}[X] = 0$ and $\text{Var}[X] = \mathbb{E}[X^2] = 0$, then $X = \mathbb{E}[X] = 0$ a.e.

X, Y independent implies

1. $f_{X,Y}(x, y) = f_X(x)f_Y(y)$
2. $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
3. $\text{Cov}[X, Y] = 0$

Gaussian:

$$f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-(x-\mu)^2/2\sigma^2}$$

$$f_{X(x_{1:N})} = \frac{\exp\left(-\frac{1}{2}(x - \mu)^\top K^{-1}(x - \mu)\right)}{(2\pi)^{N/2} \sqrt{\det K}}$$

$$K = \mathbb{E}[(x - \mu)(x - \mu)^\top]$$

$$= \begin{pmatrix} \sigma_1^2 & \text{Cov}[X_1, X_2] & \dots \\ \text{Cov}[X_2, X_1] & \sigma_2^2 & \dots \\ \vdots & \vdots & \ddots \end{pmatrix}$$

For X, Y jointly gaussian, $\text{Cov}(X, Y) = 0 \implies X, Y$ independent. Further, any linear combination of jointly gaussian random variables is also (jointly) gaussian.

Adding random variables:

$$\begin{aligned} f_{X+Y}(z) &= \int f_X(x) f_Y(z - x) dx \\ &= (f_X \star f_Y)(z) \end{aligned}$$

For X, Y independent gaussian, $X + Y$ is gaussian with mean $\mu_X + \mu_Y$ and variance $\sigma_X^2 + \sigma_Y^2$.

X, Y independent implies

1. $f_{X,Y}(x, y) = f_X(x)f_Y(y)$
2. $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$
3. $\text{Cov}[X, Y] = 0$

Random process: $X : \omega \mapsto f(t, \omega)$ (random variable for each fixed t , random function for each fixed ω)

Stationary Gaussian process:

1. Mean: $\mathbb{E}[X(t)] = 0$
2. Covariance: $\mathbb{E}[X(s)X(t)] = K(s - t)$ for some function K (i.e. depends only on the time difference)

The covariance function and covariance matrix are related by

$$\begin{aligned} K &= \mathbb{E} \left[\begin{pmatrix} X(t_1) \\ X(t_2) \end{pmatrix} \begin{pmatrix} X(t_1) & X(t_2) \end{pmatrix} \right] \\ &= \begin{pmatrix} K(0) & K(t_1 - t_2) \\ K(t_2 - t_1) & K(0) \end{pmatrix} \end{aligned}$$

with $K(t_1 - t_2) = K(t_2 - t_1)$

White Gaussian Process: $K(s - t) = N_0 \delta(s - t)$

Spectral Density: $S_X(f) = \mathcal{F}[K(\tau)]$ (Fourier transform of the covariance function)

Autocorrelation: $R_\tau = \mathbb{E}[X(t)X(t + \tau)]$ (same as covariance for zero-mean processes)

Gaussian LTIs: Let X be stationary white gaussian and $h(t)$ be the impulse response of an LTI system. Then for $Y = h \star X$,

$$\mathbb{E}[Y] = \int h(t - \tau) \mathbb{E}[X(\tau)] d\tau = 0$$

$$K_Y(\tau) = \mathbb{E}[Y(t)Y(t + \tau)]$$

$$= \mathbb{E} \left[\int \int X(\lambda) X(\sigma) h(t - \lambda) h(t + \tau - \sigma) d\lambda d\sigma \right]$$

$$= \mathbb{E} \left[\int \int K_X(\lambda - \sigma) h(t - \lambda) h(t + \tau - \sigma) d\lambda d\sigma \right]$$

$$= N_0 \int h(t - \sigma) h(t + \tau - \sigma) d\sigma$$

Cauchy-Schwarz:

$$\left(\int x(t)y(t) dt \right)^2 \leq \int |x(t)|^2 dt \cdot \int |y(t)|^2 dt$$

Energy: $\mathcal{E}_x = \int |x(t)|^2 dt$

Matched filter: $h(t) = x(T - t)$ maximizes the SNR ($|b| \mathcal{E}_x / N_0$) at time T for a signal $x(t)$ corrupted by white gaussian noise.

Equivalent to maximizing the correlation $\int (bx(t) + w(t))x(t) dt = b \int_0^T |x(t)|^2 dt + \int_0^T w(t)x(t) dt$

Probability of error:

$$\begin{aligned} P_e &= \mathbb{P}(\hat{b} = 1 | b = -1)(1 - p) \\ &\quad + \mathbb{P}(\hat{b} = -1 | b = 1)p \\ &= (1 - p) \int_D f_{Y|b=-1}(y | b = -1) dy \\ &\quad + p \int_{D^c} f_{Y|b=1}(y | b = 1) dy \end{aligned}$$

(precisely the area under the overlapping tails of the two distributions)

Decision Rule:

$$\begin{cases} \frac{p f_{R|b=1}}{(1-p) f_{R|b=-1}} > 1 \implies \hat{b} = 1 \\ \frac{p f_{R|b=1}}{(1-p) f_{R|b=-1}} < 1 \implies \hat{b} = -1 \end{cases}$$